THADOMAL SHAHANI TSEC ENGINEERING COLLEGE

SEPM

Deepa Sunil Pandey
T21, 79

Q1] System Development Life cycle (SDLC) is a structured and systematic approach used to develop information systems. It provides a step by step process for planning, designing, developing, testing and maintaining software.

SDLC mainly focuses on software development activities while project management activities are handled under the project life cycle.

Phases of SDLC:

1. Planning -
In this phase, the project idea is defined and feasibility is checked.
- Identify problem and objectives
- Conduct feasibility study
- Prepare project plan.

2. Requirement analysis -
User requirement are collected and documented clearly,
- Gather functional and non functional requirements.
- Prepare SRS document.
- Get user approval.

3. Design -
System requirements are converted into technical design.
- Prepare HLD & LLD
- Design database and UI

THADOMAL SHAHANI
TSEC
ENGINEERING COLLEGE

- Select tools and technology

4. Implementation
Actual coding of the system is done.
- Develop modules
- Follow coding standards.
- Perform unit testing.

5. testing -
System is tested to remove errors.
- Integration testing
- System testing
- Acceptance testing.

6. Deployment -
System is installed for users.
- Install in real environment
- user training
- Go live

7. Maintainance
System is updated and improved after deployment
- Fix bugs
- Improve performance
- Add new features.

Q2] A project is a temporary and unplanned activity under to develop a unique product, service or system. Examples of project include software development, website creation, mobile app development, database design, etc.

Project management is the process of planning, scheduling, organizing, monitoring and controlling project activities to achieve the desired goals within time, cost, scope and quality limits.

Characteristics of Project management:
1. Temporary nature
2. Defined scope
3. Unique goal
4. limited Resources
5. Team coordination.
6. Risk Handling
7. Planning and control
8. Quality focus.

3] PMBOK stands for Project management Body of Knowledge. It is a standard guide developed by the project management Institute (PMI) that provides best practices, methods, tools and techniques for managing projects effectively.

PMBOK is used to manage software projects from initiation to closure. It helps project managers in planning schedules, handling risks, controlling costs, and maintaining quality.

PMBOK Knowledge Areas -
1. Integration management
2. Scope management
3. Schedule management
4. Cost management
5. Quality management.

6. Resource management.
7. Communication management
8. Risk management
9. Procurement management.
10. Stakeholder management.

Q4) Explain ITPM

→ ITPM stands for Information Technology Project managen
It means applying project management principles, tools and techniques to IT and software related projects as software development, network setup, cloud systems websites, and mobile applications.

ITPM focuses on managing software projects effectively by balancing time, cost, scope, quality and technical challenges.

Objectives of ITPM-
1. Deliver software ontime.
2. Control development cost.
3. maintain product quality.
4. manage technical risks.
5. meet user requirements.

Phases of ITPM-
1. Initiation - Define project ideas and goals.
2. Planning - estimate cost, schedule and resources.
3. Execution - Develop the software system.
4. monitoring and control - Track progress and fix issues.
5. closing - deliver product and complete documentation.

RFP stands for Request for Proposal. It is a formal document prepared by a client organization to invite software companies or vendors to submit proposals for developing a software system or providing IT services.

Role of RFP-
- clearly defines software requirements.
- Invites proposals from different vendors.
- Helps compares cost, quality and technology.
- Supports fair vendor selection.
- Reduces misunderstanding between client and developer.

Contents of RFP-
1. Organization introduction.
2. Project background
3. Functional and non-functional requirements.
4. Scope of software project.
5. Budget details.
6. Time schedule
7. Technical expectations
8. Evaluation criteria.
9. Terms and conditions.
10. Proposal submission details.

Purpose -
- To get the best software solution.
- To select capable vendors.
- To save cost and time
- To ensure project requirements are fulfilled.

Q6] A feasibility study is the process of evaluating whether a proposed software project is practical, beneficial and achievable before actual development begins. It helps management decide whether to proceed with the project.

Types of feasibility -

1. Technical feasibility - checks availability of hardware software, tools and technical skills.
2. Economic feasibility - Determines whether project benefits are greater than cost.
3. Operational feasibility - checks whether users can operate and accept the system
4. Scheduling feasibility - verifies whether the project can be completed on time.
5. legal feasibility - Ensures compliance with software licenses and legal rules.

Contents -

- Problem definition
- Current system analysis
- Proposed software solution.
- Cost benefit analysis
- Risk analysis
- Time schedule
- Final recommendation.

Purpose -

- Helps management decision making
- Reduces project risk
- Saves time and cost.

- Identifies problems early.
- Improves software planning.

WBS stands for Work Breakdown Structure. It is an important project management technique used to divide a software project into smaller and manageable parts. In SEPM, WBS helps in planning, scheduling, assigning responsibilities and controlling project work effectively.

It organises the total project work into a hierarchical structure. The top level represents the complete software project, and lower levels represent modules, tasks and sub tasks, Breaking the project into smaller activities makes execution easier and more systematic.

Levels of WBS -
1. Project
2. major deliverables / modules
3. Sub deliverables
4. Activities / Tasks.

Examples of software Project WBS -
Library management system
- Requirement analysis.
- System design
- Coding
- Testing
- Deployment
- Maintainance